Shichao Sun

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Education

Sep. 2017 – Jan. 2020(expected)
Master of Engineering, Fudan University, China
Major in Computer Technology
GPA: 3.43/4.0

Sep. 2013 – Jun. 2017
Bachelor of Management, Fudan University, China
Major in Secrecy Policy and Management
GPA: 3.15/4.0

Research Publications

1. **Sun, S.** & Xie, Z. (2017). Bilstm-based models for metaphor detection. In *National CCF Conference on Natural Language Processing and Chinese Computing (NLPCC 2017)*.

Awards

2018 **Scholarship for Outstanding Students** at Fudan University.
2017 **Outstanding Paper** of CCF NLPCC 2017.
2014 – 2016 **National Scholarship of Encouragement and Support** at Fudan University.

Skills

Language: Native Chinese and fluent English.
Coding: Python, C/C++, LaTeX
Deep Learning Framework: PyTorch

Research Experience

2017 – 2018

Math Word Problems Solving

Input a math word problem(including addition, subtraction, multiplication, division, power), output the solution expression.

- Implement the baseline (Seq2Seq model), and obtain better the result than the original paper(EMNLP 2017), as a result of more careful data processing, strict rules to restrict the generation and better hyperparameter setting.
- Study the influence of different attention mechanisms containing Bahdanau attention, Luong attention, Dot attention, Minus attention, Bilinear attention. Design and implement multiway attention network motivated by considering different information from various attention mechanism.
- According to the difference between machine translation with math word problem solving, design dynamic embeddings for number token at decoding side, the result is dramatically improved, approximately 2%.
- Data Augmentation for solution expression: generating equivalent expressions to train the model by using commutative property and distributive property based on the motivation that makes the model to learn the math knowledge.
- The code of dynamic embeddings and data augmentation is available from `https://github.com/ShichaoSun/math_pade_da`.
- Expression Tree can represent the relationship among the math units better than the sequence, so we design Seq2Tree model to Top-down generate Expression Tree for math word problems.
- Code using PyTorch, one paper is under review by IJCAI 2019.

2016 – 2017

Metaphor Detection

Classify whether a sentence contains the verb metaphor(like “ignite passion”).

- Detect verb metaphor by using Bi-LSTMs with word embeddings.
- Study the influence of different granularity input(SVO, Dependency Sequence, Sentence).
- Code with Lasagne(open source in: `https://github.com/ShichaoSun/metaphor`), and the paper is published in NLPCC 2017.

2015 – 2016

Chinese Word Embedding

Adopting word2vec to train Chinese Word Embedding in dependency tree and study the properties.